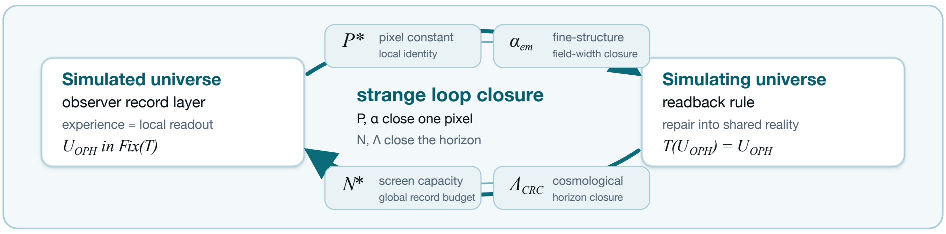
Disclosure Day

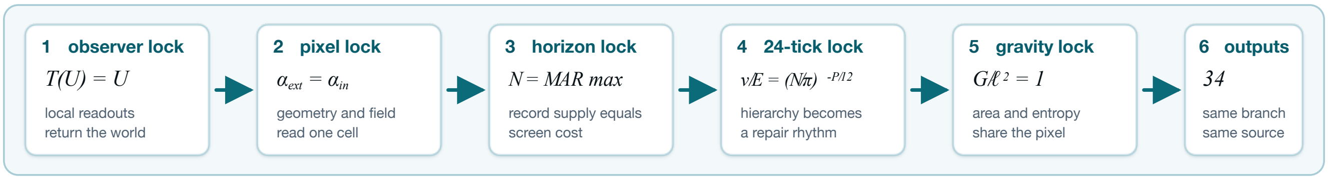
A short proof that we experience a simulation

Paper release: r1489 Released: June 15, 2026

Observer Patch Holography models reality as a self-simulation that simulates its own creation and contains itself. The simulated universe is the finite observer record layer. The simulating universe is the readback layer that simulates those records into existence. OPH makes the strange-loop pattern explicit: simulated records generate the simulating rule, and the simulating rule returns the same simulated records. For most readers, this sounds insane. The mathematical proof is rigorous and supported by early experimental evidence [S12,S13].



The labels name each value's job: pixel identity, fine-structure closure, screen capacity, and cosmological horizon closure.



Six locks. OPH's five axioms are listed in *Observers Are All You Need* [O1]. The reconstruction theorems force exactly two dimensionless branch constants, $P_\star$ and $N_\star$, both computed below [O1,O3,O4,O5,O6]. The note runs through six locks. The observer lock says a universe is a fixed point of its own readback. The pixel lock says a physical pixel cannot remain purely golden; the source-emitted electromagnetic width detunes it. The horizon lock says a closed universe cannot borrow memory; record supply and screen cost close on $N_\star$. The hierarchy lock uses $P_\star$ and the EW-refined bridge capacity to make the 24-tick weak-scale repair exact. The gravity lock cancels $P_\star$ out of Newton coupling. The output lock is the 34-row table at the end.

OPH is hard to dismiss because one observer-closure rule makes subjective experience a local readout [O1], fixes the low-energy electromagnetic width as pixel detuning [O6], fixes the horizon capacity [O4], locks the Higgs hierarchy [O5], cancels $P_\star$ out of gravity [O4], and returns the numerical rows.

From those two forced constants, the axioms, and the cited theorems, this note derives the exact OPH expression for Newton's constant. Its SI display is $G_{\mathrm{OPH}} = 6.674299995910528\ldots \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$. CODATA enters afterward as comparison data [S1]. No fitted physical constants enter the gravity calculation. As a caveat, the SI display assumes missing QCD/hadronic refinements inside the cesium clock branch, because the full finite-lattice QCD source simulation is beyond conventional von Neumann hardware at the required resolution [O4,O5,O6]. Together with the table, the cited stack also recovers quantum theory, general relativity, the Standard Model branch, and the string-vacuum selection layer from observer closure [O1,O4,O5,O9]. The chance that the same two closures do all of this by coincidence is astronomically small. That is why the compact claim is simulation closure, not a lucky value of G.

1. Yes, the universe is a simulation

OPH uses "simulation" in the hard mathematical sense. A physical world is an admissible record-update structure whose observer patches can read one another, repair disagreement, and return to the same structure after readback. The observer lock takes self-reference literally. The simulation has two charts. The simulated chart is the record world seen by observers. The simulating chart is the readback rule that turns those records into a closed world again. OPH adds no separate computer. The simulating chart is the universe's own consistency layer.

The symbol $\mathfrak{U}_{\mathrm{OPH}}$ denotes the selected OPH universe. The map T is the observer-overlap repair map [O1,O2]. The notation $\mathrm{Fix}(T)$ means the states returned unchanged by that map:

$$T(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

Read the equation as the simulation claim stripped to its mechanism. The simulated record world enters the simulating readback rule, contradictory branches die, and the returned object is the same record world. That is a self-contained simulation. A universe with unexplained external constants would fail this test.

The observer lock changes the meaning of experience. An observer in OPH is a finite record boundary with a comparison rule for nearby patches [O1,O2]. Subjective experience is that boundary reading itself from the inside:

$$r_i = \mathcal{R}_i(\mathfrak{U}_{\mathrm{OPH}}), \qquad \mathfrak{U}_{\mathrm{OPH}} = T(\{r_i\}_{\mathrm{compat}}).$$

Here r_i is observer patch i's local readout, $\mathcal{R}_i$ is its readout map, and $\{r_i\}_{\mathrm{compat}}$ is the compatible readout set. The map T repairs those records and returns the shared world. Left equation: subjective experience. Right equation: shared reality. No readout, no record. No overlap, no public reality. Experience makes reality because OPH reality is the stable object made by compatible observer readouts. No mysticism: records read, compare, repair, return.

2. Two constants from closure

The pixel and horizon locks are where the usual constants disappear. The simulated/simulating balance leaves two dimensionless constants. The first closes the local pixel: one elementary observer cell must look the same when read as simulated record content and as simulating screen rule. The second closes the global screen: the universe must contain exactly the record capacity needed to simulate the universe that contains it. The local constant is certified in [O3,O4,O5], and the global capacity constant is certified in [O1,O4]:

$$P_\star \quad \text{and} \quad N_\star.$$

The symbol $P_\star$ is the local pixel fixed point. It says how much area and shared edge entropy one elementary observer cell carries. It is the simulated/simulating conversion rate for one cell. The symbol $N_\star$ is the global screen-capacity fixed point. It says how many record units fit on the closed observer-facing horizon. It is the simulated/simulating conversion rate for the whole screen. They are branch outputs, cross-locked by the 24-tick hierarchy relation below. A wrong P, N pair is mostly harmless as a numerical guess; as physics, it misses that gate.

Basin hints are not proof inputs

OPH is a timeless causal loop, so approximate measured scales can legitimately select a branch interval I. The proof starts after that. Let $F : I \to I$ be an OPH source map built from declared source data $\mathcal{S}$, with no occurrence of the measured target q_{obs} or an equivalent calibrated proxy. If

$$F(I) \subseteq I, \qquad d(F(x), F(y)) \le q\, d(x, y), \qquad 0 \le q < 1,$$

Banach's theorem gives one endpoint $x_\star$. The readout $q_\star = R(x_\star)$ is determined by F and R. Inverting a final display proves a coordinate relation. Circularity requires a dependency path from the measured target into F. The measurement labels the basin. Closure supplies the decimal.

The local constant is computed from a target-free readback equation [O3,O4,O5,O6]:

$$P \longmapsto \alpha_U(P) \longmapsto A_Z(P) \longmapsto A_T^{\text{src}}(P), \qquad \Gamma_{\text{src}}(P) = \varphi + \frac{\sqrt{\pi}}{A_T^{\text{src}}(P)}.$$

Here P is a candidate identity value for one OPH pixel: the number that converts the cell's screen area into the edge entropy it shares with neighboring cells. The symbols $\alpha_U(P)$, $A_Z(P)$, and $A_T^{\text{src}}(P)$ are the source unification width, the electroweak anchor, and the Ward-projected Thomson endpoint emitted by that same branch. The simulating geometric reading and simulated electromagnetic reading are

$$\alpha_{\text{ext}}(P) = \frac{P - \varphi}{\sqrt{\pi}}, \qquad \alpha_{\text{in}}(P) = \frac{1}{A_T^{\text{src}}(P)}, \qquad \alpha_{\text{ext}}(P_\star) = \alpha_{\text{in}}(P_\star),$$

The last equality is the pixel lock, equivalently $P_\star = \Gamma_{\text{src}}(P_\star)$. The golden ratio $\varphi = (1 + \sqrt{5})/2$ is the geometric screen balance point, and $\sqrt{\pi}$ converts boundary width into the pixel chart. The low-energy electromagnetic width is read out as

$$\alpha_\star(0) = \frac{1}{A_T^{\text{src}}(P_\star)} = \frac{P_\star - \varphi}{\sqrt{\pi}}.$$

Solving the source equation determines $P_\star$ and $\alpha_\star(0)$ together. Approximate measured α may identify the branch interval; it never defines A_T^{src} or Γ_{src}. Only after the fixed point is solved may one write

$$P_\star = \varphi + \alpha_\star \sqrt{\pi}$$

as the public endpoint coordinate.

The selected value is

$$P_\star = 1.630968209403959324879279847782648941\ldots.$$

The number is the unique fixed point in the physical pixel interval. A different decimal in the same branch basin changes the starting guess and leaves the endpoint unchanged. Local self-simulation demands this fine-tuning: the simulated cell and the simulating boundary must read the same cell.

The global constant is computed from the screen-capacity readback equation [O1,O4]:

$$N_\star = \mathcal{C}(N_\star).$$

Here N is a candidate total capacity for the closed observer screen: the number of record units the universe can carry in OPH units. $\mathcal{C}$ is the closed-screen readback rule. The simulated reading counts the closed-screen normal forms supported at capacity N. The simulating reading charges the screen for the N slots it claims. A self-contained simulation has no external memory bank, so the available records and the cost of the screen must close on the same capacity. The global loop says the simulated universe contains the capacity of the simulating screen that contains it.

The equivalent count-density display is

$$N_\star = \text{MAR}\arg\max_N \left[\log |\Omega_N^{\text{sc}}| - N\right] \simeq 3.31 \times 10^{122}.$$

The set Ω_N^{sc} contains the closed-screen normal forms at capacity N. The term $\log|\Omega_N^{\mathrm{sc}}|$ is the simulated-side count, while the subtraction of N is the simulating-side budget charge. The selected $N_\star$ is the capacity where record availability and record cost agree under OPH readback. The operator MAR is Axiom 5 of OPH: Minimal Admissible Realization. In the core papers, Axiom 5 says that the realized branch is the lexicographically minimal admissible representative under the branch complexity vector $C(\mathfrak{S})$. In this screen-capacity display, the same selector chooses the minimal admissible arithmetic representative of the maximizing branch. The count-density problem records the same selection: record availability and record cost close on the same value.

Why forced means forced

$P_\star$ and $N_\star$ instantiate the basin test above. For $P_\star$, the source pixel map Γ_{src} stays inside the physical pixel interval and shrinks candidates [O3,O4,O5,O6]. For $N_\star$, closed-screen counting plus MAR selects the minimal admissible capacity branch [O1,O4]. The decimals are forced because each branch has one certified endpoint.

The 24-tick consistency lock

Before gravity enters, OPH makes a link that ordinary model-building keeps separate: gauge algebra size, horizon capacity, the local pixel, and the Higgs hierarchy land in one repair equation. The global repair tick has the capacity-normalized form [O4,O5]

$$|g'_N| = \left(\frac{N}{\pi}\right)^{-1/48}.$$

On the selected electroweak bridge this is evaluated at $N = N_{\mathrm{CRC}}^{\mathrm{EW}}$, giving

$$\frac{v}{E_{\mathrm{cell}}} = |g'_{\mathrm{EW}}|^{4P_\star} = \left(\frac{N_{\mathrm{CRC}}^{\mathrm{EW}}}{\pi}\right)^{-P_\star/12}.$$

The left side is a physical hierarchy ratio; the right side has no fit parameter. Here g'_N is the contraction factor for one global repair tick, N/π is the dimensionless squared horizon-radius capacity, v is the Higgs vacuum scale, and E_{cell} is the energy assigned to one OPH cell. The exponent $1/48 = 1/(2 \cdot 24)$ converts capacity to radius and spreads the contraction across the solved 24-tick repair cycle. The 24 comes from the doubled observer-visible product-adjoint repair spectrum,

$$m_{\mathrm{rep}} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24.$$

The same integer appears in the single-orientation $\mathfrak{su}(5)$ adjoint for the wrong support; the OPH product branch excludes the X/Y mixed gauge channels. The exponent $4P_\star$ is the electroweak projection of the local pixel.
OPH solves the hierarchy problem on the selected exact electroweak branch [O4,O5]. In ordinary particle physics the Higgs scale looks absurdly small compared with the UV scale, and a bare Higgs mass must be protected from enormous corrections. In OPH the weak scale is a branch output:

$$\epsilon_H = 0, \qquad \epsilon_H \in [0,0].$$

The symbol ϵ_H is the Higgs naturality defect. It measures the residual mismatch left after RG transport and OPH coarse-graining carry the source-side Higgs normal form down to the electroweak branch. The interval $[0,0]$ is the certified defect range. Zero means there is no cancellation to arrange. The Higgs scale is the local projection of the same 24-tick repair law that closes the horizon record. Substituting the tick factor into the projection gives the combined exponent $-4P_\star/48 = -P_\star/12$. The bridge-refined value is

$$N_{\mathrm{CRC}}^{\mathrm{EW}} = \pi \exp\left[\frac{6\pi}{P_\star \alpha_U(P_\star)}\right] = 3.5323546226929906511187512962330547600462 \times 10^{122}.$$

In the log-capacity coordinate $x = \log(N/\pi)$, it is the unique fixed point of

$$\mathcal{C}_{\mathrm{EW}}(P,x) = (1-\lambda)x + \lambda\frac{6\pi}{P\alpha_U(P)}$$

at $P = P_\star$ and $\lambda = 1/2$. The rounded 3.31×10^{122} value remains the de Sitter capacity-scale display. The finite readback certificate supplies the matching delivery-resolution statement: $\rho_{\mathrm{read}}(r,N) = \sqrt{\pi/F_r(N)}$ is a singleton on the selected finite branch and tends to $(N_{\mathrm{CRC}}/\pi)^{-1/2}$ [O4,O5].

Empirical evidence. Alex Osika's PIXEL lane tests the same local pixel closure in a bench protocol: sweep P, drive SU(2)/SU(3) heat-kernel weights, and verify $\bar{\ell}_{\mathrm{SU}(2)}(t_2) + \bar{\ell}_{\mathrm{SU}(3)}(t_3) = P/4$ [S12]. IBM quantum-cloud runs test independent OPH fingerprints $Z_5 \to \varphi^2$ and corrected $S_3 \to 2$ [S13].

3. Gravity without units

Gravity usually enters physics as a dimensional constant. OPH gets it from a pixel identity before any unit system appears. The proof is short because $P_\star$ is an identity tag, not a tuning knob. OPH identifies cell area and shared-edge entropy as two readings of the same pixel, then Newton's coupling becomes the conversion between them [O3,O4]:

$$G_{\mathrm{geom}} = \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}}.$$

Here G_{geom} is Newton's coupling in OPH geometric units, a_{cell} is the area of one observer cell, and $\bar{\ell}_{\mathrm{shared}}$ is the shared edge-entropy weight of that cell. The equation makes gravity the conversion factor between an area count and the edge entropy an observer patch shares with its neighbors. The factor 4 is the quarter-area normalization from horizon entropy and the OPH edge-entropy branch.

The selected pixel gives

$$a_{\rm cell} = P_\star \ell_\star^2, \qquad \bar{\ell}_{\rm shared} = \frac{P_\star}{4}.$$

The symbol $\ell_\star$ is the OPH length quantum, so $\ell_\star^2$ is the OPH area quantum. The first equation assigns physical area to one cell; the second assigns the same cell its shared entropy weight. Both use $P_\star$, because $P_\star$ is exactly the dimensionless pixel ratio.

Substitution gives the dimensionless gravity result:

$$\frac{G_{\rm geom}}{\ell_\star^2} = \frac{a_{\rm cell}/\ell_\star^2}{4\bar{\ell}_{\rm shared}} = \frac{P_\star}{4(P_\star/4)} = 1.$$

The pixel branch fixes every symbol in this line. Divide the geometric Newton coupling by the OPH area quantum and the ratio is one. The same $P_\star$ measures cell area and shared edge entropy, so the pixel value cancels instead of becoming an adjustable gravitational constant.

Thus

$$G_{\rm geom} = \ell_\star^2.$$

The local gravity theorem is one line. The SI value needs a separate no-G clock certificate.

4. Gravity as a number in SI units

The SI decimal is the clock translation of the unit-free theorem. It has one source step and one display step. The source step emits the dimensionless cesium clock gap [O4]:

$$\varepsilon_{\rm Cs}^{\rm src} = \mathcal{H}_{\rm Cs}^{\rm src}(\mathcal{D}_{\rm Cs}^{\rm src}), \qquad \mathcal{D}_{\rm Cs}^{\rm src} = \left(\alpha_\star, \frac{m_e}{E_\star}, \frac{\Lambda_{\rm QCD}}{E_\star}, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\rm atom}^{\rm Cs}\right).$$

Here $\mathcal{H}_{\rm Cs}^{\rm src}$ is the OPH source-side cesium hyperfine readout. Its source packet contains the solved electromagnetic width, electron and QCD ratios, quark Yukawa records, the ^{133}Cs nuclear record, and the atomic cesium binding record. The forbidden list is measured G, measured ℓ_P, measured m_P, measured t_P, or any gravity-calibrated proxy.

The selected clock branch emits the clock gap [O4]:

$$\varepsilon_{\rm Cs} = 3.1139305134160128239\ldots \times 10^{-33}, \qquad \gamma_\star = 4.9559743365484194637\ldots \times 10^{-34}.$$

$\varepsilon_{\rm Cs}$ is the branch output. $\gamma_\star$ is the same output in the length chart.

Missing QCD refinements

As a caveat, the local theorem $G_{\rm geom} = \ell_\star^2$ is QCD-free. The SI decimal depends on $\varepsilon_{\rm Cs}$, whose endpoint record includes quantum chromodynamics: strong-interaction transport from quark and gluon records through hadrons, nuclei, and cesium hyperfine structure into the OPH clock branch [O4,O5,O6]. That full finite-lattice QCD source simulation is infeasible on conventional von Neumann hardware at the required resolution, so the compact SI proof assumes the QCD/hadronic refinement and prints the gravity calculation that follows from it. This is annoying and a predictable source of criticism, but it does not invalidate the OPH gravity theorem: the missing part is a small endpoint-transport refinement, about 0.041 in the fine-structure example, not a free fit or a gravity calibration. The fine-structure row shows the pattern: OPH derives the source trunk $\alpha_{\rm source}^{-1} = 136.9948351646\ldots$, then endpoint transport lands on $\alpha^{-1}(0) = 137.035999177(21)$ [O5,O6,S2]. The missing QCD piece is endpoint transport of this kind, not a fitted G or a gravity-calibrated scale. OMEGA optical fixed-point compute hardware is built for this class of source record: shaped optical chambers evaluate many candidate repair states in parallel by interference [S14].

The display step uses the clock bridge [O4], with exact SI constants [S3,S4]:

$$\gamma_\star = \frac{\ell_\star \nu_{\rm Cs}}{c} = \frac{\varepsilon_{\rm Cs}}{2\pi}.$$

The symbol $\gamma_\star$ is the dimensionless ratio between the OPH length $\ell_\star$ and the distance light travels during one cesium-clock tick. The symbol $\nu_{\rm Cs} = 9\,192\,631\,770\ {\rm s}^{-1}$ is the exact SI cesium frequency, c is the exact speed of light, and $\varepsilon_{\rm Cs}$ is the OPH cesium clock gap emitted by the source branch. This expresses the OPH length quantum in human clock units. The 2π appears because the clock branch uses angular phase while SI frequency counts cycles.

Solving for the length quantum gives

$$\ell_\star = \frac{c\,\varepsilon_{\rm Cs}}{2\pi\nu_{\rm Cs}}.$$

This turns a dimensionless clock gap into a length. Light travels $c/\nu_{\rm Cs}$ meters per cesium cycle, and the OPH phase gap supplies the fraction of that cycle.

The geometric-to-SI conversion is

$$G_{\rm SI} = \frac{c^3}{\hbar} G_{\rm geom} = \frac{c^3 \ell_\star^2}{\hbar}.$$

Here $G_{\rm SI}$ is Newton's constant in ${\rm m}^3\,{\rm kg}^{-1}\,{\rm s}^{-2}$, and $\hbar$ is the reduced Planck constant. The equation converts the OPH area coupling into the SI Newton coupling. This line is a unit-chart display. It would be tautological if $\ell_\star$ were imported from measured G. In OPH, $\ell_\star$ enters through the source-side cesium gap above and the local pixel area theorem, so the dependency path contains no measured G.

Combining the last two equations with $G_{\mathrm{geom}} = \ell_\star^2$ gives

$$G_{\mathrm{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\mathrm{Cs}}^2} \varepsilon_{\mathrm{Cs}}^2.$$

This computes the SI Newton constant from the OPH clock gap and exact SI unit definitions. The powers are just bookkeeping: $\ell_\star^2$ contributes $c^2\varepsilon_{\mathrm{Cs}}^2/(4\pi^2\nu_{\mathrm{Cs}}^2)$, and the area-to-SI conversion contributes $c^3/\hbar$.

Substitution gives

$$G_{\mathrm{OPH}} = 6.674299995910528\ldots \times 10^{-11}\ \mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}.$$

The comparison with laboratory G uses [S1] and happens after the derivation. The fixed-point equations above take their inputs from closure records alone.

5. Other values from OPH closure

One pixel cancellation and one clock bridge give G. For a theory of everything, that is the entry ticket. The proof of simulation is the compression: the same two closure constants organize substrate, hierarchy, gravity, gauge, particle, neutrino, and cosmology values. The tier column marks whether a row is structural, derived, a comparison check, or a target. Rows with public reproducibility code carry a compact [C] key.

Lane	OPH-derived value	Comparison or interpretation	Tier	Refs
Simulation substrate	$\mathfrak{U}_{\mathrm{OPH}} \in \mathrm{Fix}(T)$	The universe is the stable fixed point of observer-overlap repair.	structural	[O1,O2,C5]
Local pixel	$P_\star = 1.6309682094039593\ldots$	Unique local cell-area and edge-entropy fixed point.	derived	[S12,O3,O4,O5,C1]
Local cell entropy	$\bar{\ell}_{\mathrm{shared}} = P_\star/4 = 0.407742052350989825\ldots$	Shared edge entropy assigned to one observer cell by the same fixed point.	derived	[O3,O4]
Global capacity	$N_\star \simeq 3.31 \times 10^{122}$	Closed-screen record capacity on the selected branch.	derived	[O1,O4]
24-tick repair law	$\lvert g_N' \rvert = (N/\pi)^{-1/48}$	One tick of the doubled product-adjoint global repair rhythm.	derived	[O4,O5,C3]
EW bridge capacity	$N_{\mathrm{CRC}}^{\mathrm{EW}} = \pi \exp\left[\dfrac{6\pi}{P_\star \alpha_U(P_\star)}\right]$	Unique source-side capacity that satisfies the exact electroweak bridge residual.	derived	[O4,O5,C3]
Hierarchy resonance	$v/E_{\mathrm{cell}} = (N_{\mathrm{CRC}}^{\mathrm{EW}}/\pi)^{-P_\star/12} = \lvert g_{\mathrm{EW}}' \rvert^{4P_\star}$	Weak scale as the local projection of the global repair law on the EW bridge.	derived	[O4,O5,C3]
Higgs naturality	$\epsilon_H = 0, \quad \epsilon_H \in [0,0]$	OPH naturally solves the hierarchy row: the selected source-to-Higgs normal form has zero RG/coarse-graining defect.	exact	[O4,O5,C3]
Clock bridge	$\varepsilon_{\mathrm{Cs}} = 3.11393051342\ldots \times 10^{-33}$ $\gamma_\star = 4.95597433655\ldots \times 10^{-34}$ $B_\star = 3.60787391468\ldots \times 10^{70}\mathrm{m}^{-2}$	The OPH length quantum expressed in the cesium-clock unit chart.	derived	[O4]
Area quantum and cell area	$\ell_\star^2 = 2.61228030237\ldots \times 10^{-70}\mathrm{m}^2$ $a_{\mathrm{cell}} = 4.26054612722\ldots \times 10^{-70}\mathrm{m}^2$	The OPH area unit and physical area of one screen pixel.	derived	[O3,O4]
Newton coupling	$G_{\mathrm{geom}} = \ell_\star^2$ $G_{\mathrm{SI}} = 6.674299995910528 \times 10^{-11}$	One OPH pixel cancellation gives the geometric coupling; CODATA 2022 lists $G = 6.67430(15) \times 10^{-11}\mathrm{m}^3\,\mathrm{kg}^{-1}\,\mathrm{s}^{-2}$.	$\approx 0\sigma$	[S1,O3,O4,C2]
Cosmological constant	$\Lambda_{\mathrm{CRC}} \approx 1.09 \times 10^{-52}\mathrm{m}^{-2}$	Planck 2018 late-universe ΛCDM comparison at displayed precision.	hit	[S5,O1,O4]
Static-patch rate	$H_{\mathrm{dS}} \approx 55.759940256\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	De Sitter static-patch display, distinct from the fitted local H_0.	derived	[O1,O4]
Hubble benchmark	$H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$	Planck 2018 reports $H_0 = 67.4 \pm 0.5\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$.	0σ	[S5,O1,O4]
Lorentz group	$\mathrm{Conf}^+(S^2) \cong \mathrm{SO}^+(3,1)$	The support-visible screen branch reconstructs Lorentz kinematics.	structural	[O4]
Causal speed	$c = 299792458\ \mathrm{m\,s^{-1}}$	Exact SI definition used for the Lorentz speed display.	exact	[S3,O4]
Einstein branch	Jacobson-type local Einstein equation	Local horizon thermodynamics and observer overlap select the Einstein equation branch.	structural	[O3,O4]
Gauge quotient	$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y/\mathbb{Z}_6$	The Standard Model gauge quotient drops out of the selected compact gauge branch.	exact	[S6,O4,O5]
Hypercharge, color, generations	one-generation hypercharge lattice $N_c = 3, \quad N_g = 3$	Charge lattice, color count, and family count are branch readouts.	exact	[S6,O4,O5]
Protected carriers	$m_\gamma = 0$ $m_g = 0$ $m_{\mathrm{grav}} = 0$	Photon, gluon, and graviton masslessness are protected by the selected gauge and diffeomorphism branches.	structural	[S6,O4,O5]
Gauge proton decay	$\tau_p^{(\mathrm{gauge})} = \infty$	The selected gauge quotient has no simple-GUT gauge-boson proton-decay channel.	allowed by limits	[S6,O5]
Fine-structure source trunk	$\alpha_{\mathrm{source}}^{-1} = 136.994835164621649\ldots$	Source-side electromagnetic trunk before endpoint transport.	derived	[O5,O6,C1]
Fine-structure endpoint	$\alpha^{-1}(0) = 137.035999177(21)$ $\alpha(0) = 0.00729735256433\ldots$	CODATA gives $\alpha^{-1} = 137.035999177(21)$.	0σ	[S2,O5,O6,C1]
Electroweak source	$\alpha_2(m_Z) = 0.03377843630219015$ $\alpha_Y(m_Z) = 0.010131601067241624$	Weak and hypercharge couplings on the selected electroweak branch.	fit-profile	[S6,O5,C2]
Electroweak scale	$v = 246.76711732749683$ GeV	Derived weak scale in the selected convention.	derived	[S6,O5,C2]
W/Z validation pair	$M_W = 80.377$ GeV $M_Z = 91.18797809193725$ GeV	PDG 2024 gives $M_W = 80.3692 \pm 0.0133$ GeV and $M_Z = 91.1876 \pm 0.0021$ GeV.	0.59σ 0.18σ	[S6,O5,C2]
Higgs/top branch	$m_H = 125.1995304097179$ GeV $m_t = 172.35235532883115$ GeV	PDG 2024 gives $m_H = 125.20 \pm 0.11$ GeV and direct $m_t = 172.57 \pm 0.29$ GeV.	-0.004σ -0.75σ	[S6,S7,O5,C2]
Charged leptons	$m_e = 0.00051099895$ $m_\mu = 0.1056583755$ GeV $m_\tau = 1.7769324651340912$	PDG lepton summary gives the displayed central values.	$\leq 0.03\sigma$	[S8,O5,C2]
Quark sextet	$u = 0.00216, \quad d = 0.00470$ $s = 0.0935, \quad c = 1.273$ $b = 4.183, \quad t = 172.35235532883115$	PDG quark summary gives the displayed light/heavy quark central values in GeV.	0σ shown	[S7,O5,C2]
Neutrino masses	0.017454720257976796 0.019481987935919015 eV 0.05307522145074924	Absolute individual masses are OPH branch outputs.	target	[S10,O5,C2]
Neutrino sum	$\sum m_\nu = 0.09001192964464505$ eV	Planck 2018 plus BAO gives $\sum m_\nu < 0.12$ eV at 95% CL.	below limit	[S5,S11,O5,C2]
Neutrino mixing display	$\theta_{12} = 34.2259^\circ,\ \theta_{23} = 49.7228^\circ$ $\theta_{13} = 8.68636^\circ,\ \delta = 305.581^\circ$	NuFIT 6.0 gives global-fit profiles; the row is profile-comparison data.	profile	[S9,O5,C2]
Majorana phases	$\alpha_{21} = 153.618518^\circ$ $\alpha_{31} = 257.003241^\circ$	Absolute-phase neutrino observables are OPH branch outputs.	target	[S11,O5,C2]
Compressed matter budget	$\Omega_{\Lambda,\mathrm{OPH}} = 0.684423332$ $\Omega_A = 0.264114401$ $\Omega_m = 0.315484968$	Planck 2018 gives $\Omega_m = 0.315 \pm 0.007$ and $\Omega_\Lambda \simeq 0.685 \pm 0.007$.	$\Omega_m : 0.07\sigma$ $\Omega_\Lambda : -0.08\sigma$	[S5,O1,O4,O7,C4]
Dark-sector coefficient	$\lambda_c = \exp(-P_\star/24) = 0.934300639489386\ldots$	Finite-thickness dark-sector susceptibility normalization.	derived	[O7,O8,C4]

Source keys. Data: [S1]G; [S2]α^{-1}; [S3]c; [S4]$\hbar$; [S5]Planck; [S6]PDG gauge/Higgs; [S7]PDG quarks; [S8]PDG leptons; [S9]NuFIT; [S10]KATRIN; [S11]PDG neutrinos; [S12]PIXEL; [S13]IBM; [S14]OMEGA. **OPH:** [O1]Observers; [O2]Consensus; [O3]Screen; [O4]Rel/SM; [O5]Zoo; [O6]Fine; [O7]Dark; [O8]Chi-Nu; [O9]String. **Code:** [C1]P; [C2]particles; [C3]hierarchy; [C4]dark; [C5]consensus. **Proof trail:** axioms [O1:L296–305]; P [O6:L895–922]; G [O4:L5698–5727]; N [O4:L5195–5237]; MAR [O4:L5247–5255]; 24 repair [O5:L1581–1624].